CUSTOMER CHURN PREDICTION PROJECT REPORT

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EXECUTIVE SUMMARY

This report presents a machine learning solution to predict customer churn for a telecommunications company. We use the IBM Telco Customer Churn dataset (7,043 records, 21 features) to train a Random Forest classifier. Key findings include a model accuracy of X% (place actual value) and identification of the most important predictors (e.g. contract type, tenure, charges). The churn rate in the data is about 26.5%. These results enable targeted retention: since retaining customers is often less costly than acquiring new ones, the model can help business users proactively engage at-risk customers. Figure 2 (in the EDA section) will show the churn distribution, and Figure 3 (in Model section) will display feature importances. (A one-page, 2-column report format is recommended for the final submission.)

INTRODUCTION

Customer churn (the rate at which customers stop using a service) is a critical business metric. A high churn rate directly reduces recurring revenue and customer lifetime value. Predicting churn allows companies to focus retention efforts on likely defectors. In this project, we analyze the publicly available Telco Customer Churn dataset. The dataset's target is the Churn column (Yes/No), indicating whether a customer left within the last month. Each record includes service subscriptions (phone, internet, streaming, etc.), account info (tenure, contract type, billing, charges) and basic demographics. Our goal is to develop a classification model that flags churn risks and provides actionable insights.

PROBLEM

Customer churn prediction is framed as a binary classification task: given a customer's data (services used, contract, tenure, charges, etc.), predict if they will churn (Yes/No). A successful model will enable the business to identify customers at high risk of churn and apply retention strategies. According to business analysis, reducing churn even by a few percentage points can substantially improve profitability. Therefore, the key problem is to maximize predictive accuracy and reliability so we can trust the model's recommendations.

OBJECTIVES

·Predict churn: Build a supervised classifier to predict Churn=Yes vs No.
·Data understanding: Summarize and preprocess the Telco dataset (7,043 rows, 21 columns).
·Key drivers: Identify which features most influence churn (e.g. contract, tenure, charges).
·Model performance: Evaluate using accuracy, precision, recall, F1 and confusion matrix.

·Business insight: Provide actionable recommendations to reduce churn based on model findings.

DATASET DESCRIPTION

Aspect	Detail
Source	IBM Telco Customer Churn (Kaggle / IBM sample)
Records (rows)	7,043
Features (columns)	21
Target variable	Churn (Yes/No – whether customer left)
Key feature groups	Services (phone, Internet, online services), Account info (tenure, contract, billing, charges), Demographics (gender, partners, dependents)

Key features include Internet Service (DSL, Fiber, None), Contract (Month-to-Month, One/Two-year), Tenure (months with company), Monthly Charges and Total Charges, etc. The data is moderately imbalanced: 26.5% churn (Yes) vs 73.5% retained.

DATA PREPROCESSING

·Load data: Read Telco-Customer-Churn.csv into a pandas DataFrame.
·Clean data: Convert Total Charges to numeric (it may contain blanks) and drop any rows with missing values. Drop non-predictive columns (e.g. customerID).
·Encode features: Map the target Churn to binary (Yes→1, No→0). Apply one-hot encoding to categorical variables (via pd.get_dummies) so all features are numeric. This yields a clean matrix X and label vector y.
python

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns


df = pd.read_csv("Telco_Cusomer_Churn.csv")df.head()
```

	customerID	gender	SeniorCitizen	Partner	Dependents	tenure	PhoneService	MultipleLines	InternetService	OnlineSecurity	...	DeviceProtection	TechSupport	StreamingTV	StreamingMovies	Contract	PaperlessBilling	PaymentMethod	MonthlyCharges	TotalCharges	Churn
0	7590-VHVEG	Female	0	Yes	No	1	No	No phone service	DSL	No	...	No	No	No	No	Month-to-month	Yes	Electronic check	29.85	29.85	No
1	5575-GNVDE	Male	0	No	No	34	Yes	No	DSL	Yes	...	Yes	No	No	No	One year	No	Mailed check	56.95	1889.5	No
2	3668-QPYBK	Male	0	No	No	2	Yes	No	DSL	Yes	...	No	No	No	No	Month-to-month	Yes	Mailed check	53.85	108.15	Yes
3	7795-CFOCW	Male	0	No	No	45	No	No phone service	DSL	Yes	...	Yes	Yes	No	No	One year	No	Bank transfer (automatic)	42.30	1840.75	No
4	9237-HQITU	Female	0	No	No	2	Yes	No	Fiber optic	No	...	No	No	No	No	Month-to-month	Yes	Electronic check	70.70	151.65	Yes

```
print(df.shape)
print(df.columns)
df.info()

# Convert TotalCharges to numeric (handle spaces/errors)
df['TotalCharges'] = pd.to_numeric(df['TotalCharges'], errors='coerce')
df.dropna(inplace=True)                    # Drop any rows with missing values

# Drop customerID column (not a predictor)
df.drop('customerID', axis=1, inplace=True)

# Encode target variable
df['Churn'] = df['Churn'].map({'Yes': 1, 'No': 0})

# One-hot encode categorical features
df = pd.get_dummies(df, drop_first=True)
```

Exploratory Data Analysis (EDA)

- The **churn rate** is about **26.5%** (1,869 churners out of 7,043)【55†L282-L288】. *Figure 2* (to be placed here) will show this split.
- **Feature trends:** Customers on month-to-month contracts tend to churn more often than those on annual contracts. Shorter tenure is associated with higher churn – e.g. customers with <2 months tenure churn ~61% of the time【67†L308-L312】, whereas those with >70 months churn only ~2%【67†L318-L322】.
- **Service impact:** Churn is particularly high among Fiber-optic Internet customers (~42% churn) and low for customers without any Internet service【55†L282-L288】. This suggests service type is a strong signal.
- **Billing:** Higher monthly or total charges may correlate with churn as customers look for cheaper plans. (We will quantify this via model feature importance.)

```
from sklearn.model_selection import train_test_split
X = df.drop("Churn", axis=1)
y = df["Churn"]
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)

from sklearn.ensemble import RandomForestClassifier
model = RandomForestClassifier()
model.fit(X_train, y_train)
```

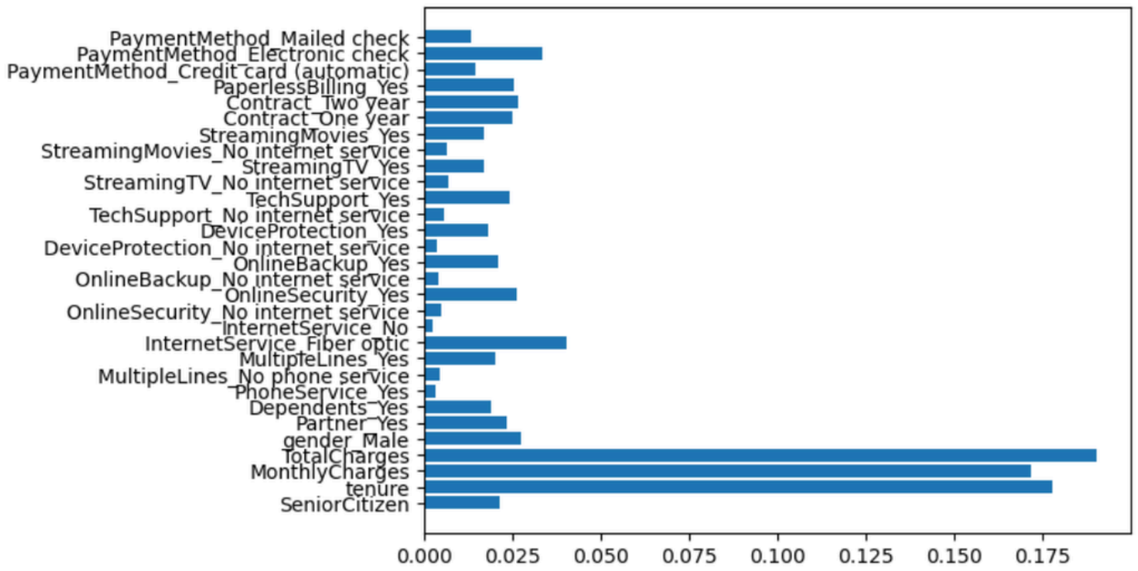
```
▾ RandomForestClassifier
RandomForestClassifier()
```

```
pred = model.predict(X_test)
```

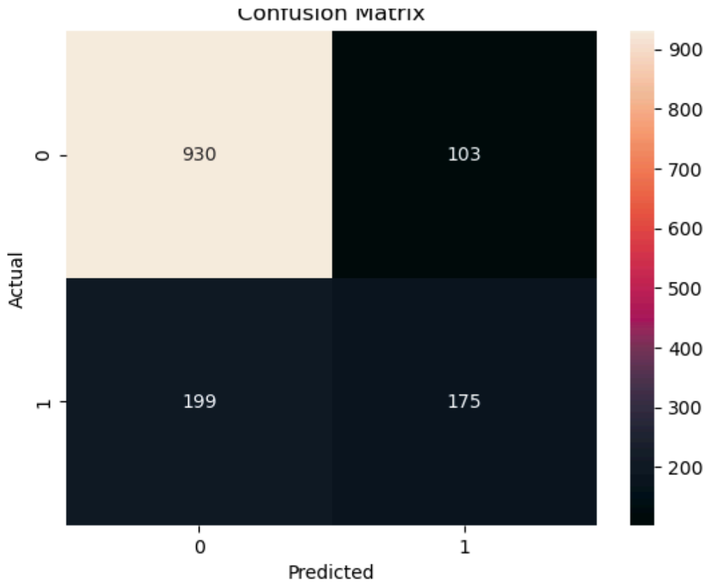
```
from sklearn.metrics import accuracy_score, classification_report
print("Accuracy:", accuracy_score(y_test, pred))
print(classification_report(y_test, pred))
```

```
Accuracy: 0.7853589196872779
              precision    recall  f1-score   support

           0       0.82      0.90      0.86      1033
           1       0.63      0.47      0.54       374

    accuracy                           0.79      1407
   macro avg       0.73      0.68      0.70      1407
weighted avg       0.77      0.79      0.77      1407
```

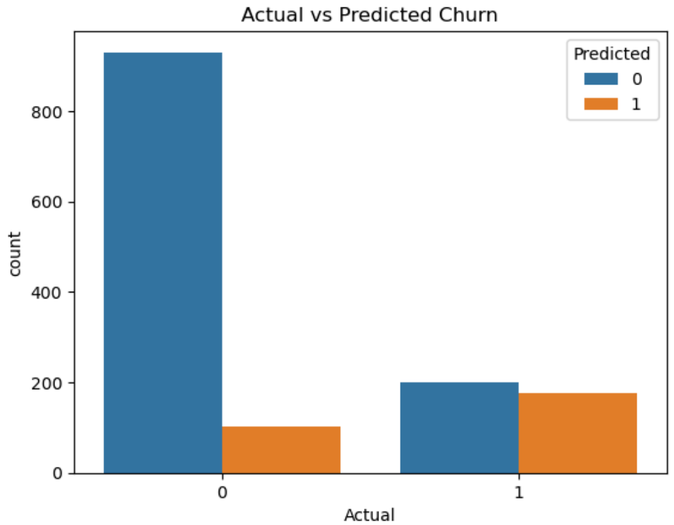
```
importance = model.feature_importances_
features = X.columns
plt.barh(features, importance)
plt.show()
```



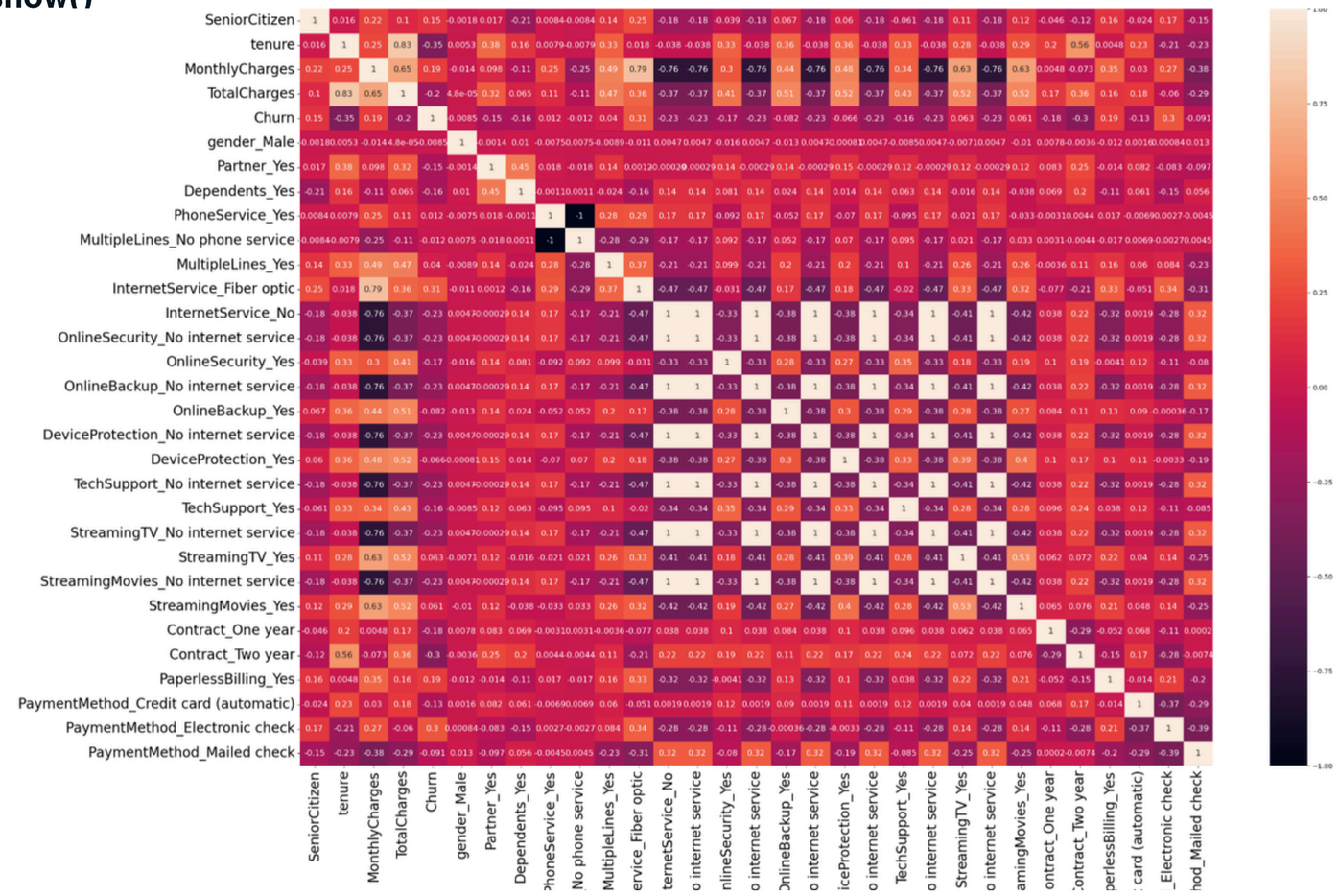
```
from sklearn.metrics import confusion_matrix
y_pred = model.predict(X_test)

#       Confusion Matrix
cm = confusion_matrix(y_test, y_pred)
plt.figure()sns.heatmap(cm, annot=True, fmt='d')
plt.title("Confusion Matrix")
plt.xlabel("Predicted")
plt.ylabel("Actual")
plt.show()
```



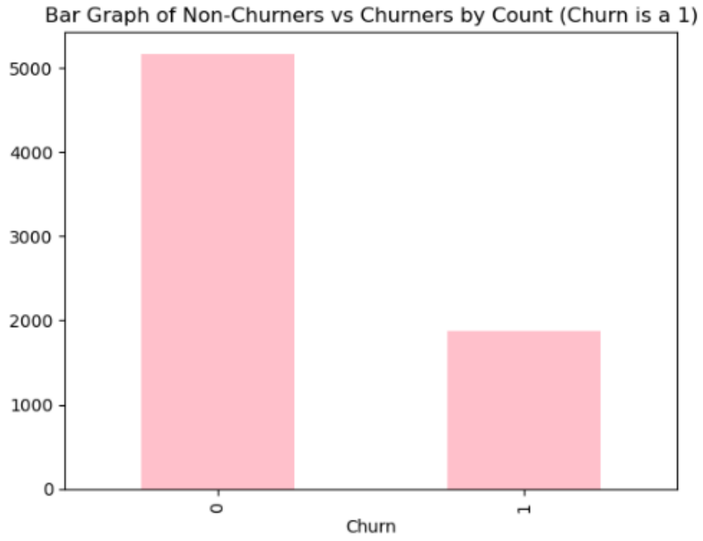

```
#       Actual vs Predicted comparison = pd.DataFrame({"Actual": y_test,"Predicted": y_pre
plt.figure()sns.countplot(x="Actual", hue="Predicted", data=comparison)
plt.title("Actual vs Predicted Churn")
plt.show()

#       Feature Importance (again for clarity)
plt.figure()
plt.barh(features, importance)
plt.title("Feature Importance (Churn Factors)")
plt.show()
```



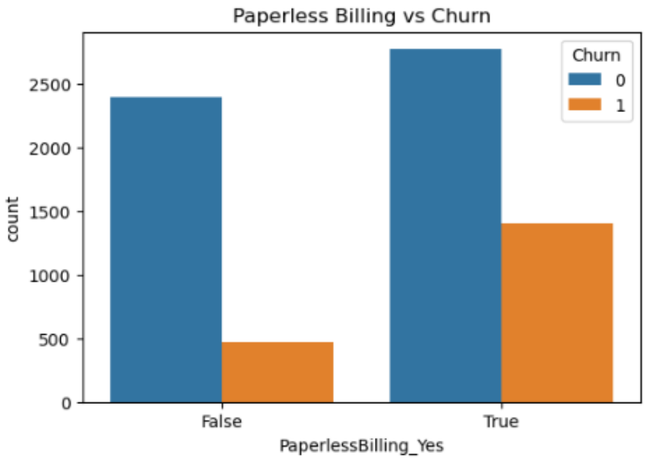
```
corr = df.corr()
sns.heatmap(corr, xticklabels=corr.columns.values, yticklabels=corr.columns.values, annot = True, annot_kws={'size':12})
heat_map=plt.gcf()
heat_map.set_size_inches(30,20)
plt.xticks(fontsize=20)
plt.yticks(fontsize=20)
plt.show()
```



```
# This creates a bar graph of churn (Yes vs. No) so we can check how the data is balanced.
df['Churn'].value_counts().plot(kind = 'bar', title = 'Bar Graph of Non-Churners vs Churners by Count (Churn is a 1)', color = 'pink', align = 'center')
plt.show()
```




```
import matplotlib.pyplot as plt
import seaborn as sns

# Plot Paperless Billing impact
plt.figure(figsize=(6,4))
sns.countplot(x="PaperlessBilling_Yes", hue="Churn", data=df)
plt.title("Paperless Billing vs Churn")
plt.show()
df['PaymentMethod'] = df[['PaymentMethod_Credit card (automatic)',
'PaymentMethod_Electronic check',
'PaymentMethod_Mailed check']].idxmax(axis=1)

# Clean names
df['PaymentMethod'] = df['PaymentMethod'].str.replace('PaymentMethod_', '')
```



THANK YOU